

Tide retrospective: the window convergence dominated convergence for the numerator remainder

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The analytic heart of the numerator expansion: on the mesoscopic window, the normalized remainder integral tends to zero,

$$\int_{W_q} P(z) \frac{e^{-(L(qz)-L(0))/q^2} - e^{-T_2(z)} \sum_{j \leq N} P_j(z) q^j}{q^N} dz \longrightarrow 0 \quad (q \rightarrow 0^+),$$

by the filter-indexed dominated convergence theorem with the window indicator folded into the integrand, exactly as the architecture consult prescribed. The dominator is the previous tide's q -uniform majorant times the observable's growth; the pointwise limit at fixed z is the scalar graded expansion, activated by the vanishing scaled remainder and switched on by eventual window membership. The tide also defines the expansion's coefficient integrals $a_j = \int P e^{-T_2} P_j$ and proves them integrable. The two outer tails and the final packaging were deliberately cut to the next tide: they are a full tide of set-integral comparisons, and a clean cut beats a half-done state.

1 Setting

Stage 5c-ii-a of the forward-expansion programme (the consult's item 7, with items 8–9 deferred). The consult's DCT guidance was followed literally: integral over all of $\mathbb{R}^d$ with the varying window as an indicator (set integrals with a varying set leave one unfolding restrictions manually), dominated convergence via the filter-indexed theorem, and measurability in z for each fixed q only.

2 The thing formalised

`boltzmann_factor_eq` extracts the majorant tide's factoring of the true Boltzmann factor through the exponent split as a standalone lemma (used by both the majorant and the pointwise limit here). `numeratorCoeff` and `integrable_coeff_integrand` give the coefficient integrals, dominated by $(1 + \|z\|)^{n+j+2N} e^{-(\lambda/2)\|z\|^2}$ through the quadratic lower bound and the 5a growth. `tendsto_integral_window_remainder` is the dominated convergence: eventual measurability mirrors the seabed's rescaled-DCT block (the loss is only measurable, everything else continuous); eventual domination is the 5c-i majorant times the observable bound $|P| \leq 2C_P(1 + \|z\|)^n$, with the q^N cancelling the normalization; the dominator integrates by the binomial-Gaussian helper; and the pointwise limit composes `exp_graded_expansion`

(at $a_s = V_s(z)$, ρ the scaled remainder) with `IsLittle0.tendsto_div_nhds_zero`, a constant multiplication, and an eventual-equality transfer through `boltzmann_factor_eq` and eventual window membership.

3 Where progress stalled

Two rounds. Rewriting the target with $0 = \int 0$ also rewrote the zero inside the filter $\mathcal{N}[>](0)$ — state the DCT conclusion with $\int 0$ and transfer by `simpα` instead of rewriting the goal. And positivity cannot see $0 < \lambda/4$ for an opaque domain field (pass `linarith [D.lambda_pos]`). A third, self-inflicted round: a scripted re-indentation of the four DCT bullets mangled the nested case split; rewriting the block wholesale was faster than repairing it.

4 Mathlib lookup versus new mathematics

Lookup: `tendsto_integral_filter_of_dominated_convergence`, `AEStronglyMeasurable.indicator`, `Set.indicator_of_mem / notMem`, `Integrable.mono'`, `IsLittle0.tendsto_div_nhds_zero`, `Tendsto.congr'`. New: none — the stage was built so that this proof would be composition, and it was (one build iteration of substance).

5 What the next tide inherits

Stage 5c-ii-b, the last piece of the numerator stage: the observable-weighted outer tail of the true integrand (binomial split into `integrand_meso_tail_isLittle0`), the coefficient-polynomial outer tail (`gaussian_meso_tail_isLittle0` with the 5a growth), the eventual decomposition identity splitting the numerator into window piece plus tails, and the packaging of `numerator_hasExpansion` as a `Laplace.IsAsymptoticExpansionTo`. After that: stage 6 (asymptotic division by the partition function, whose constant coefficient $a_0 = \int e^{-T_2} > 0$) and stage 7 (the public forward existence and jet-comparison theorems).